0.9

0 → 0.9 × 2 = 1.8 → 1
0.8 × 2 = 1.6 → 1
0.6 × 2 = 1.2 → 1
0.2 × 2 = 0.4 → 0
0.4 × 2 = 0.8 → 0
0.8 × 2 = 1.6 → 1

$0.9 = 0.11100_2$ Binary | 0.111 00 1100 1100 11001 001
$= 0.71463_8$ Octal | 0.111 0011 001100
$= 0.E6$ Hex |

$0.11100_2 \times 2^0$

= 01110011 | 00110011 | 00110011 | 0000 0000
NASA Hex → 7 3 3 3 3 3 0 0

0.11100×2^0
1.1100×2^{-1}

1111111
− 1
1111110

IEEE 754 → 00111111 | 0110 0110 | 0110 0110 | 0110 0110
3 F 6 6 6 6 6 6

5. 75

$.75 \times 2 = 1.5 \rightarrow 1$

0101 $\quad .5 \times 2 = 1.0 \rightarrow 1$

so $5.75_{10} = 0101.1100_2$ Binary

$= 05 . 6_8$ Octal

$= 5 . C_{16}$ Hex

0101.1100×2^0

$= 0.1011100 \times 2^3$

$= 0.1011100 | 00000000 | 00000000 | 00000011$

NASA Hex → 5 C 0 0 0 0 0 3

$= 101.1100 \times 2^0$

$= 1.011100 \times 2^2$

$$\begin{array}{r} 1111111 \\ + \quad 10 \\ \hline 10000001 \end{array}$$

$\underbrace{01000000}_{\text{Power}} | 10111000 | 00000000 | 0000 0000$

IEE754 → 4 0 B 8 0 0 0 0

99.7

$99/16 = 6$ reminder 3

$6/16 = 0$ reminder 6

so $99_{10} = 63_{16} = 0110\ 0011_2$

$.7 \times 2 = 1.4 \rightarrow 1$

$.4 \times 2 = 0.8 \rightarrow 0$

$.8 \times 2 = 1.6 \rightarrow 1$

$.6 \times 2 = 1.2 \rightarrow 1$

$.2 \times 2 = 0.4 \rightarrow 0$

$.4 \times 2 = 0.8 \rightarrow 0$

repeating

so $99.7_{10} = 0110\ 0011.1\underline{0110}_2$ Binary

$= 001\ 100\ 011.101\ 100\ 110\ 011\ 001\ 100\ 110$

$= 1\ 4\ 3.5\ \underline{4\ 6\ 3\ 1}_8$ Octal

$= 0110\ 0011.1011\ 0011\ 0011\ 0011$

$= 6\ 3.B\ \underline{3}$ Hex

$= 0110\ 0011.1\underline{0110}_2 \times 2^0$

$= 0.110\ 0011\ 1\ \underline{0110}_2 \times 2^7$

$= 0110\ 0011 \mid 1011\ 0011 \mid 0011\ 0011 \mid 0000\ 0111$

NASA Hex $\rightarrow$ 6 3 B 3 3 3 0 7

$110\ 0011.1\ 0110 \times 2^0$

$1.10\ 0011\ 1\ \underline{0110} \times 2^6$

$$\begin{array}{r} 1111111 \\ +\quad 110 \\ \hline 10000101 \end{array}$$

$= 0100\ 0010 \mid 1\ 1\ 000\ 111 \mid 0110\ 0110 \mid 0110\ 0110$

4 2 C 7 6 6 6 6

3) a) $5.75 = 101.11_2$ S S.F

can fit in 8 bits - 3 bits for 101 = 5 bits (2^5)

$5.75_{10} \times 2^5 = 101.11_2 \times 2^5$

$= 10111000_2$ shifted 5 bits

b) $0.9 = 0.1\underline{1100}_2$ will ~~use~~ max bits

scale to 2^{16}

$0.9_{10} \times 2^{16} = 0.\underline{11100}_2 \times 2^{16} = 184$

$= 1110011001100110_2$ shifted 16 bits

c) $99.7 = 1100011.1\underline{0110}_2$ will use max bits

can fit to 24 bits - 7 bits for 1100011 = 17 bits (2^{17}

$99.7 \times 2^{17} = 1100011.1\underline{0110} \times 2^{17}$

$= 110001110110\ 0110\ 0110\ 0110_2$ shifted 17 bits

4) a) $5.75_{10} \times 2^5 = 184$

$184 \times 7 = 1288$

shift back: 1288×2^{-5}

$\downarrow$

$1288 / 16 = 80$ reminder 8

$80 / 16 = 5$ reminder 0

so $1288_{10} \times 2^{-5} = 08_{16} \times 2^{-5}$

$= 00001000_2 \times 2^{-5}$

$= 0.01000_2$ $\left(\simeq 40.25_{10} \simeq 5.75_{10} \times 7_{10}\right)$

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b) $0.9_{10} \times 2^{16} \approx 58982$

$58982 \times 7 = 412874$

shift back 412874×2^{-16}

$\downarrow$

$412874 / 16 = 25804$ reminder 10 (A)

$25804 / 16 = 1612$ reminder 12 (C)

$1612 / 16 = 100$ reminder 12 (C)

$100 / 16 = 6$ reminder 4

$6 / 16 = 0$ reminder 6

so $412874_{10} \times 2^{-16} = 64CCA_{16} \times 2^{-16}$

$= 0110\ \underline{0100}\ \underline{1100}\ \underline{1100}\ \underline{1010}_2 \times 2^{-16}$

$= 0110.0100110011001010_2$ $\left(\simeq 6.3_{10} = 0.9_{10} \times 7_{10}\right)$

c) $99.7_{10} \times 2^{17} = 9147514 6_{10}$

$91475146_{10} \times 7 = 640326022$

$640326022 / 16$	$= 40020376$		.375
$40020376 / 16$	$= 2501273$	reminder 8	.5
$2501273 / 16$	$= 156329$	~ 9	.5625
$156329 / 16$	$= 9770$	~ 9	.5625
$9770 / 16$	$= 610$	~ 10 (A)	.625
$610 / 16$	$= 38$	~ 2	.125
$38 / 16$	$= 2$	~ 6	.375
$2 / 16$	$= 0$	reminder 2	

$91475146_{10} \times 7 = 640326022_{10}$

$= 262A998_{16}$

$= 0010\ 0110\ 0010\ 1010\ 1001\ 1001\ 1000_2$

shift back: $0010\ 0110\ 0010\ 1010\ 1001\ 1001\ 1000_2 \times 2^{-17}$

$= 100110001.0101\ 01001100\ 1100_2$

$\simeq 697 \simeq 99.7 \times 7 \simeq 697.9$